Resolution Semantics for Partial Algebras

A Lean-Checked Many-Sorted Encoding Bridge and a Binary Observational Completion

Adrian Puha

17 August 2026

Abstract

Starting from Diaconescu's total-algebra encoding of partial algebras, we give a concrete Lean-checked bridge for many-sorted signatures with arbitrary finite arities and disjoint total and partial operation families. The bridge implements the Horn theory Γ, the model reduct β, the sentence translation α, and a canonical free model γ. It checks the two term-evaluation lemmas, the α–β satisfaction condition, recovery of the source partial algebra up to canonical equivalence, and the unique-lift property of γ. We also check the functor laws, package the lift as a hom-set equivalence, and derive preservation and reflection of semantic consequence for arbitrary theories of closed formulas. This part is positioned as a machine-checked continuation in a concrete Lean representation, not as a novelty claim for the underlying encoding.

In the binary Resolution layer, we formalize a normalized free compatible completion of binary partial algebras. Old-defined applications reduce to old values; old-undefined applications become structured finite-tree Answers. The Answer algebra has the expected universal property and is the expression algebra modulo equality after resolution.

Our additional construction is an intrinsic finite-complement separating semantics. Every distinct pair is distinguished by a compatible total extension that fixes the old carrier pointwise and whose complement injects into $\mathrm{Fin}(n+1)$. At each budget, pair separation by this intrinsic class is equivalent to pair separation by the canonical finite-tag models

$$D.\mathrm{Carrier} \ \sqcup \ \mathrm{Fin}(n) \ \sqcup \ \{\star\}.$$

The fresh-tag count is at most the combined constructor size; this is a construction bound, not an optimality claim. When the old carrier is finitely coded, the same witnesses are finite in the ordinary sense, so the generated Answer algebra is residually finite with an explicit total-state bound. For a finite operation signature, the qualitative residual-finiteness conclusion also follows from classical recognizability of finite ground-equation congruence classes; the construction here additionally keeps the old carrier faithful and supplies the displayed bound.

Finite-tag agreement induces a separated filtration whose Cauchy completion supports canonical operations and has exactly the same universal equations in the language with named old-carrier constants. For a finitely coded old carrier, the embedding is surjective exactly when the old evaluator is total. Without any finiteness assumption, an undefined old application plus a context $x \mapsto u(x, e)$ fixing every old element yields a factorial Cauchy orbit with no generated limit. Every finite stage is strictly coarser than equality, and least separating budgets are unbounded. Natural and integer arithmetic satisfy this criterion using $0/0$ and $x \mapsto x + 0$, so both completion embeddings are proper. Division by zero remains a structured non-old Answer, not an ordinary number. Lean 4.33.0 checks the development. The exact priority of this combined package remains under specialist review.

Contents

1 Introduction **3**

2 A Lean-Checked Bridge to Diaconescu's Encoding **5**

2.1 The encoded models, β, and α . . . 5

2.2 The canonical model γ and its lift . . . 5

2.3 Functor laws and semantic consequence . . . 6

3 Generated Answers as a Free Compatible Completion **7**

3.1 Expressions and Answers . . . 7

3.2 Universal property . . . 8

3.3 Quotient presentation . . . 8

4 Intrinsic Finite-Complement Separation **9**

4.1 Pair-dependent separator . . . 10

5 The Observational Filtration and Completion **12**

5.1 Relative universal property . . . 14

6 Exact Equational Conservativity **15**

7 Arithmetic Examples **16**

7.1 Natural numbers . . . 16

7.2 Integers . . . 16

8 Relation to Prior Work **17**

8.1 Free completion and partial algebras . . . 17

8.2 Ground terms, recognizability, and operation trees . . . 17

8.3 Tree metrics and completion . . . 17

8.4 Finite complements and profinite comparison . . . 18

8.5 Totalized division . . . 19

8.6 Evidence-qualified contribution . . . 19

9 Formalization and Trust Boundary **19**

10 Limitations and Open Questions **21**

11 Conclusion **21**

1 Introduction

The paper has two deliberately separated scopes. First, it connects the Resolution normal-form construction directly to Diaconescu's many-sorted encoding of partial algebras as total algebras [18]. This bridge allows arbitrary finite operation arities and distinguishes designated total symbols from partial symbols. Second, the separation, filtration, completion, properness, and arithmetic developments specialize to the one-sorted binary setting used by the original Resolution project. Claims below state explicitly which of these two scopes is in force.

Let D be a one-sorted binary partial algebra with carrier A, operation symbols Σ, and evaluator

$$\mathrm{eval}_D : \Sigma \longrightarrow A \longrightarrow A \longrightarrow \mathrm{Option}(A).$$

The value $\mathrm{some}(c)$ records that the old application is defined and equals c; none records old undefinedness. We call A *finitely coded* when it admits an injection into $\mathrm{Fin}(m)$ for some m. This is classically equivalent to finiteness and is the exact hypothesis used by the Lean statements below.

There is extensive prior work on completing or encoding partial algebras. Climent Vidal and Cosme Llópez prove that the inclusion of total algebras into partial algebras has a left adjoint, the absolutely free completion functor, and develop the functorial Schmidt construction [28]. Diaconescu gives the Horn-style encoding of partial algebras as total algebras on which our first bridge is based [18]. Completion questions also occur in partial combinatory algebra and term-rewriting models [7, 8, 23]. Accordingly, this paper does not claim that the existence of a free or total extension is new. Classical tree-automata results also imply ordinary residual finiteness for the finite-base, finite-signature ground-equational presentation discussed below [14, 20, 27]; that qualitative special case is not claimed as new either.

Our starting presentation is operationally simple. Resolve an expression bottom-up. If a node has two old operands and the old evaluator returns a value, replace the node by that old value. Otherwise retain the node, its operation symbol, and its resolved children. The resulting finite trees are called *Answers*. This gives a normalized representative for a free compatible total extension.

After checking that encoding bridge, the binary Resolution layer asks two narrower questions.

(Q1) Can equality of generated Answers be characterized by total models that do not themselves contain Answer trees, while preserving every old-defined application?

(Q2) Can those finite observations support a completion on which the operations remain canonical and all constant-bearing universal equations are unchanged?

The answer to both is yes. The paper therefore separates one bridge result from two principal binary Resolution results.

Theorem 1.1 (Many-sorted encoding bridge)**.** *Let $\Sigma = (S, TF, PF)$ be a many-sorted partial-algebra signature whose operations have arbitrary finite arities. In the concrete operations-only Lean presentation developed here:*

(a) *the Horn clauses Γ define a reduct β and the translation α satisfies $M \models \alpha(\rho)$ if and only if $\beta(M) \models \rho$;*

(b) *every partial* Σ*-algebra* D *has a canonical encoded model* $\gamma(D)$ *satisfying* Γ*, and* $\beta(\gamma(D))$ *is canonically equivalent to* D *as a partial algebra;*
(c) *every partial homomorphism* $D \to \beta(M)$ *has a unique encoded lift* $\gamma(D) \to M$ *agreeing with it under that equivalence.*
(d) *the induced maps on homomorphisms respect identities and composition, the lift is an explicit hom-set equivalence, and for arbitrary sets* E, E' *of closed formulas,*

$$E \models_{PA} E' \iff \alpha(E) \cup \Gamma \models_{ALG} \alpha(E').$$

Theorem 1.2 (Structured free completion and finite separation)**.** *For every one-sorted binary partial algebra* D*:*

(a) *the generated Answer algebra* $R(D)$ *is initial among compatible total algebras equipped with a map from the old carrier;*
(b) $R(D)$ *is represented by the quotient of the absolutely free expression algebra by the kernel of the resolution evaluator;*
(c) *every distinct* $x, y \in R(D)$ *is separated by a compatible total extension whose complement over* A *has at most* $|x| + |y| + 1$ *points; equivalently, a canonical finite-tag model needs at most* $|x| + |y|$ *fresh tags and one overflow state. Consequently, when* A *is finitely coded,* $R(D)$ *is residually finite.*

Theorem 1.3 (Observational completion and equation preservation)**.** *Finite-tag agreement defines a separated descending filtration on* $R(D)$*. Its Cauchy completion* $\widehat{R(D)}$ *is complete, contains an injective stage-dense image of* $R(D)$*, and carries the unique jointly nonexpansive extensions of the primitive operations. Relative to complete, separated filtered compatible total extensions whose old embedding is injective, the completed algebra has the expected universal extension property. If the old carrier is finitely coded, the canonical embedding is surjective exactly when the old evaluator is total; equivalently, the completion is proper exactly when some old application is undefined. Without assuming finite coding of the old carrier, suppose some old application is undefined and there are* $u \in \Sigma$ *and* $e \in A$ *with* $\mathrm{eval}_D(u, a, e) = \mathrm{some}(a)$ *for every* $a \in A$*. Then every finite observation stage is strictly coarser than equality, an explicit factorial orbit is Cauchy with no generated limit, its separation ranks are unbounded, and the completion embedding is not surjective. Moreover, for every pair of constant-bearing algebraic terms* s, t*,*

$$\widehat{R(D)} \models s = t \iff R(D) \models s = t.$$

The first two clauses of theorem 1.2 place the generated construction within established free-completion theory. The more specific ingredients are the intrinsic finite-complement observer class over a pointwise-preserved base, its pairwise separator, and the observational structure it induces. The linear estimate is retained as a general construction bound rather than a state-complexity advance. Over a finitely coded base it also gives an explicit, base-faithful bound on the entire separating target; ordinary residual finiteness itself is classical in the finite-signature subcase. The finite-base properness criterion explains exactly when the Cauchy construction adds genuinely new points over a finitely coded old carrier; the old-fixing-context criterion shows that properness is not confined to finite bases. The two hypotheses are incomparable: the one-point algebra is finitely coded and partial but has no old-fixing context, whereas natural arithmetic has an old-fixing context but no finite code for its carrier. Natural and integer arithmetic are concrete corollaries. Theorem 1.1 is a checked continuation of Diaconescu's construction and is not included in the novelty claim. We have not found the exact separation–completion package in the literature reviewed so far, but this is not a claim of historical priority.

2 A Lean-Checked Bridge to Diaconescu's Encoding

This section records the exact connection to [18]. It is intentionally phrased as a bridge and specialization; the underlying encoding results and their priority remain with the cited work. Diaconescu works with a many-sorted signature $\Sigma = (S, TF, PF)$, where symbols in TF are total and symbols in PF are partial. The Lean structure assigns each symbol a natural-number arity, a sort at every finite argument position, and a result sort. Neither the set of sorts nor either symbol family is required to be finite.

2.1 The encoded models, β, and α

An encoded total model M consists of a data carrier M_s for each $s \in S$, an auxiliary truth carrier M_b, total interpretations of all TF and PF symbols on the data carriers, operations

$$e_s : M_s \times M_s \longrightarrow M_b, \qquad \mathsf{true} \in M_b,$$

and the Horn conditions Γ. The implemented clauses say that total symbols preserve and reflect definedness, partial symbols reflect definedness, and

$$x\ e_s\ y = \mathsf{true} \quad \Longrightarrow \quad x\ e_s\ x = \mathsf{true} \quad \text{and} \quad x = y.$$

Thus the carrier of the reduct is

$$\beta(M)_s = \{x \in M_s \mid x\ e_s\ x = \mathsf{true}\}.$$

Designated-total operations restrict to total operations on these subtypes; a PF symbol is defined precisely when its totalized result is marked as defined. Encoded homomorphisms induce partial-algebra homomorphisms between the corresponding reducts.

The formula syntax is indexed by a list of bound-variable sorts. Typed de Bruijn variables make quantifier extension explicit. It is generated by existence-equality atoms, conjunction, negation, and one-variable universal quantification; iterating the last constructor represents a finite sorted variable block. The translation sends

$$t \overset{e}{=} t' \quad \longmapsto \quad (t\ e_s\ t' = \mathsf{true}),$$

commutes with the Boolean constructors, and guards a total universal quantifier by definedness of the newly bound variable.

The two term lemmas corresponding to Lemmas 3.3 and 3.4 of [18] are `Resolution.Diaconescu.ManySorted.term_evalPartial_sound` and `Resolution.Diaconescu.ManySorted.term_evalPartial_complete`. Structural induction on formulas, with the guard handling the quantifier case, then gives `Resolution.Diaconescu.ManySorted.alpha_satisfaction_condition`. The theorem is stated for assignments and hence also yields the closed-sentence instance in theorem 1.1(a).

2.2 The canonical model γ and its lift

For a partial algebra D, raw values at each sort are either old values or formal applications. The normalizer treats the two symbol families differently:

- a TF application whose arguments are old always reduces to the old total result;
- a PF application whose arguments are old reduces exactly when the source partial evaluator returns a value;
- every remaining application is retained as a sorted formal suspension with its normalized arguments.

The generated subtype consists of raw values having a term witness. Its interpreter into any compatible total target exists and is unique. This is the many-sorted finite-arity free-completion argument used for the data sorts of $\gamma(D)$.

The auxiliary carrier of $\gamma(D)$ has one true element and a failure value for each sorted pair that is not the same diagonal old value. Existence equality returns true exactly on that old diagonal. Lean checks that this model satisfies all clauses of Γ in `Resolution.Diaconescu.ManySorted.gamma_satisfiesGamma`. The old embedding is bijective onto the defined part, and it respects both the total and partial evaluators; this is packaged by `Resolution.Diaconescu.ManySorted.betaGammaPartialAlgEquiv`. Finally, `Resolution.Diaconescu.ManySorted.gamma_persistent_liberality` constructs the lift of every $D \to \beta(M)$ and proves uniqueness from the free interpreter on data and the homomorphism laws on the truth carrier.

2.3 Functor laws and semantic consequence

The continuation module makes the categorical content explicit while keeping the formal dependency surface small. It proves the identity and composition laws for β, packages persistent liberality as the checked bijection

$$\mathrm{Hom}_{ALG_\Gamma}(\gamma(D), M) \cong \mathrm{Hom}_{PA}(D, \beta(M)),$$

and defines the induced action of γ on homomorphisms. The latter respects identities and composition. These statements are exposed by `Resolution.Diaconescu.ManySorted.betaHom_comp`, `Resolution.Diaconescu.ManySorted.gammaBetaHomEquiv`, and `Resolution.Diaconescu.ManySorted.gammaHom_comp`.

Because Lean represents $\beta(\gamma(D))$ as canonically equivalent to D rather than definitionally equal to it, reflection of consequence requires an explicit transport result. The theorem `Resolution.Diaconescu.ManySorted.Formula.sat_iff_of_equiv` proves invariance of satisfaction under partial-algebra equivalence, including the negation and sorted-quantifier cases. Combining this with the α–β satisfaction condition and the canonical γ model gives `Resolution.Diaconescu.ManySorted.semantic_consequence_equivalence`, the concrete Lean form of Corollary 4.3 of [18].

The data carrier is exposed as the many-sorted Resolution normal-form type. Theorems `Resolution.Diaconescu.ManySorted.gammaHom_generatedOld`, `Resolution.Diaconescu.ManySorted.gammaHom_generatedTotal`, and `Resolution.Diaconescu.ManySorted.gammaHom_generatedPartial` show that the functorial action preserves old values and the two generated operation constructors. This is the explicit interface between the established encoding and the Resolution normal-form layer used here.

There are two deliberate scope boundaries. The Lean theorem uses canonical partial-algebra equivalence rather than identifying $\beta(\gamma(D))$ and D by definitional equality. It does not mechanize the surrounding institution/comorphism infrastructure, arbitrary signature signature morphisms, the initial-model construction of Corollary 4.2, or the proof-theoretic Birkhoff completeness result of Section 5. These boundaries are why the result is described here as a many-sorted encoding bridge and not as a replacement for the broader theory of [18].

3 Generated Answers as a Free Compatible Completion

3.1 Expressions and Answers

Finite input expressions and raw Answers are defined by separate grammars:

$$e ::= \mathrm{val}(a) \mid \mathrm{app}(f, e_1, e_2),$$

$$r ::= \mathrm{old}(a) \mid \mathrm{susp}(f, r_1, r_2).$$

The separation is intentional. Expressions are input syntax. Answers are normal forms produced by the partial evaluator together with the suspension rule.

Definition 3.1 (One-step resolution)**.** For $f \in \Sigma$ and raw Answers x, y, let

$$\mathrm{liftOp}_D(f, x, y) = \begin{cases} \mathrm{old}(c), & x = \mathrm{old}(a),\ y = \mathrm{old}(b),\ \mathrm{eval}_D(f, a, b) = \mathrm{some}(c), \\ \mathrm{susp}(f, x, y), & \text{otherwise.} \end{cases}$$

The resolution function Res_D is the structural recursion determined by

$$\mathrm{Res}_D(\mathrm{val}(a)) = \mathrm{old}(a),$$

$$\mathrm{Res}_D(\mathrm{app}(f, e_1, e_2)) = \mathrm{liftOp}_D(f, \mathrm{Res}_D(e_1), \mathrm{Res}_D(e_2)).$$

The generated carrier is the image of this evaluator:

$$R(D) = \{r \mid \exists e,\ \mathrm{Res}_D(e) = r\}.$$

The old embedding is $\eta_0(a) = \mathrm{old}(a)$. Generated operations are

$$f_{R(D)}(x, y) = \mathrm{liftOp}_D(f, x, y).$$

Generation is closed under these operations because expression witnesses can be joined by one application node.

Proposition 3.2 (Conservativity on defined applications)**.** *If* $\mathrm{eval}_D(f, a, b) = \mathrm{some}(c)$, *then*

$$f_{R(D)}(\eta_0(a), \eta_0(b)) = \eta_0(c).$$

The map η_0 *is injective.*

The proposition is immediate from definition 3.1. Its role is not depth but control: no old-defined calculation is reinterpreted.

3.2 Universal property

A *compatible total algebra over D* consists of a total Σ-algebra T, a map $i : A \to T$, and the preservation law

$$\mathrm{eval}_D(f, a, b) = \mathrm{some}(c) \quad \Longrightarrow \quad f_T(i(a), i(b)) = i(c).$$

The map i is not required to be injective. This is the target notion needed for the ordinary free-completion universal property.

Every raw Answer has an interpretation in T:

$$\Phi_T(\mathrm{old}(a)) = i(a), \qquad \Phi_T(\mathrm{susp}(f, x, y)) = f_T(\Phi_T(x), \Phi_T(y)).$$

The preservation law gives

$$\Phi_T(\mathrm{liftOp}_D(f, x, y)) = f_T(\Phi_T(x), \Phi_T(y)).$$

The only non-structural case is the old-defined branch, where the equation is exactly the compatibility condition.

Theorem 3.3 (Free compatible completion)**.** *For every compatible total algebra T over D, there is a unique homomorphism*

$$\Phi_T : R(D) \longrightarrow T$$

such that $\Phi_T \circ \eta_0 = i$.

Proof. Existence is the fold above. It preserves generated operations by the one-step equation. For uniqueness, choose an expression e generating a given Answer. Induction on e forces every compatible homomorphism to agree with the fold on $\mathrm{Res}_D(e)$. Since every generated Answer has such a witness, the two maps agree everywhere. □

The Lean-facing theorem is `ResolutionSemantics.freeCompletionUniversal`. This result confirms that the generated construction belongs to free-completion theory rather than constituting a new category-theoretic notion.

3.3 Quotient presentation

Define an equivalence on input expressions by

$$e \sim_D e' \quad \Longleftrightarrow \quad \mathrm{Res}_D(e) = \mathrm{Res}_D(e').$$

Then resolution descends to a map

$$\overline{\mathrm{Res}}_D : \mathrm{Expr}(D)/{\sim_D} \longrightarrow R(D).$$

Proposition 3.4 (Kernel quotient presentation)**.** *The map $\overline{\mathrm{Res}}_D$ is injective and surjective. Thus generated Answers are a normalized presentation of the expression algebra modulo the resolution kernel.*

The two inverse laws are formalized by `ResolutionSemantics.expressionKernelQuotientInjective` and `ResolutionSemantics.expressionKernelQuotientSurjective`.

Remark 3.5 (Relation to the Schmidt construction). The Schmidt construction and the left adjoint free-completion functor are direct prior art [28]. Our formal theorem proves the corresponding universal property for the concrete one-sorted binary presentation used here. A fully explicit categorical isomorphism between the Lean object and every formulation of the many-sorted Schmidt construction is not needed for the results below and is not claimed. The correct positioning is that $R(D)$ is a normalized free compatible completion.

4 Intrinsic Finite-Complement Separation

The free-completion theorem alone does not show that distinctions between suspended Answers are visible in independently specified models. The primitive observer condition should not depend on a particular sum encoding.

Let T be a compatible total extension with injective old embedding $i : A \hookrightarrow T$. Define its genuine outside part by

$$\mathrm{Out}_D(T) = \{z \in T \mid \forall a \in A,\ z \neq i(a)\}.$$

Definition 4.1 (Intrinsic finite-complement observer)**.** An observer has budget n when there is an injection

$$\mathrm{Out}_D(T) \hookrightarrow \mathrm{Fin}(n+1).$$

Thus only the complement over the pointwise-preserved base is finite; the old carrier itself may be infinite. The budget is an upper bound, so no ordering, chosen tags, or distinguished overflow point is part of this definition.

Every compatible extension has the carrier decomposition

$$T \cong A \sqcup \mathrm{Out}_D(T).$$

For the concrete observer used in the construction below, the outside part is canonically $\mathrm{Fin}(n) \sqcup \{\star\}$.

Definition 4.2 (Finite-tag model)**.** For $n \in \mathbb{N}$, a finite-tag model over D has carrier

$$A \sqcup \mathrm{Fin}(n) \sqcup \{\star\}$$

and total operations preserving every old-defined application:

$$\mathrm{eval}_D(f, a, b) = \mathrm{some}(c) \Longrightarrow f_T(\mathrm{old}(a), \mathrm{old}(b)) = \mathrm{old}(c).$$

No further equations are imposed.

The formal representation theorem proves that intrinsic separation at budget n is equivalent to finite-tag separation at the same budget. In the forward direction, an intrinsic complement code yields a base-fixing injection into the canonical carrier. Unused tag states are allowed and are sent to overflow when the operations are transported. This is an injection presentation, not a claimed bijection with all canonical states.

Write $x \equiv_n y$ when every finite-tag model with n tags interprets x and y equally. Full finite observational agreement means

$$x \equiv_{\mathrm{fin}} y \quad \Longleftrightarrow \quad \forall n,\ x \equiv_n y.$$

4.1 Pair-dependent separator

Let $x \neq y$ be generated Answers. Form a finite list $S(x, y)$ containing all subterms of the two raw trees. Repetition is harmless; first-occurrence indices provide a finite set of tags. Define an encoding on selected nodes:

$$q(\mathrm{old}(a)) = \mathrm{old}(a), \qquad q(s) = \mathrm{tag}(\mathrm{index}(s))$$

for selected suspended nodes s. The encoding is injective on the selected closure.

The model operation first preserves an old-defined application. Otherwise it attempts to decode both inputs, constructs one suspended node, and re-encodes that node if it belongs to the selected closure; all unrelated cases go to $\star$. There is no branch conflict. A generated normal Answer cannot contain a suspended node with two old children whose old application is defined, because resolution would already have reduced it.

Induction on selected subterms gives

$$\Phi_T(s) = q(s).$$

In particular, $\Phi_T(x) = q(x)$ and $\Phi_T(y) = q(y)$, so injectivity of q on $S(x, y)$ separates the roots.

Theorem 4.3 (Intrinsic finite-complement separation). *For distinct $x, y \in R(D)$ there exist n and a compatible total extension T whose outside part injects into* $\mathrm{Fin}(n + 1)$ *and such that*

$$\Phi_T(x) \neq \Phi_T(y).$$

One may choose

$$n = |x| + |y|,$$

where $|-|$ is raw constructor count. Consequently,

$$x \equiv_{\mathrm{fin}} y \iff x = y.$$

The intrinsic public Lean declarations are `ResolutionSemantics.qualitativeFiniteComplementSeparating_theorem`, `ResolutionSemantics.intrinsicSeparationIffFiniteTag`, and `ResolutionSemantics.intrinsicExtensionHasCanonicalPresentation`. The canonical finite-tag consequences remain exposed as `ResolutionSemantics.finiteExternalSeparation`, `ResolutionSemantics.finiteObservationComplete`, and `ResolutionSemantics.finiteSeparationBound`.

Remark 4.4 (Interpretation of the result). The old carrier A may be infinite. The theorem is therefore not finite residuality of the whole carrier. It says intrinsically that a chosen pair requires only finitely many *new* semantic states beyond A; the finite-tag sum is a derived normal form. The displayed linear estimate is a baseline construction bound, not an optimality theorem. We avoid using "full abstraction" as a novelty label; the mathematical content is the explicit model class and separator.

More precisely, let $\mathcal{C}_D$ be the class of compatible total extensions that fix A pointwise and have finite complement over A. The theorem says that $R(D)$ is residually $\mathcal{C}_D$: every unequal pair is separated by its canonical homomorphism into some member of $\mathcal{C}_D$. This is an exact classification of the statement as a relative residual property, not a claim that the terminology or general residual paradigm is new.

Corollary 4.5 (Finite-base residual finiteness)**.** *Suppose the old carrier admits an injective code into* $\mathrm{Fin}(m)$*. Then* $R(D)$ *is residually finite in the ordinary homomorphic sense. More precisely, distinct* x, y *are separated by a finite compatible algebra with an injective code into*

$$\mathrm{Fin}(m + |x| + |y| + 1).$$

The formal statements are `ResolutionSemantics.ResidualComparison.finiteBaseCompatibleSeparationBound` and `ResolutionSemantics.ResidualComparison.finiteBaseGeneratedResiduallyFinite`. The target produced by the proof still embeds the old carrier injectively, so the formal result is slightly stronger than ordinary residual finiteness, whose separating homomorphisms need not be faithful on the chosen generators.

Remark 4.6 (Finite-signature ground-equation comparison)*.* Assume in addition that the operation signature is finite. Regard each $a \in A$ as a constant and let E_D consist of the finitely many ground equations

$$f(a, b) = c \qquad \text{for every } \operatorname{eval}_D(f, a, b) = \operatorname{some}(c).$$

Every expression is congruent modulo E_D to a term presenting its resolved Answer. Conversely, every equation in E_D is respected by resolution. Thus equality after resolution is exactly the congruence generated by E_D.

Kozen proved that the classes of a finitely generated term congruence are recognizable tree languages [20]. For one such class, its finite-index syntactic congruence contains the defining congruence: congruent terms remain congruent in every context and therefore have the same membership in that class. The syntactic quotient separates the chosen class from every different one, so ordinary residual finiteness in this finite-base, finite-signature subcase follows from that classical result. Foundational regular-tree systems, finite ground rewriting, standard tree-automata treatments, and many-sorted recognizability theorems provide the broader context [9, 14, 16, 27]. We therefore make no novelty claim for that qualitative consequence. The explicit construction above retains the old carrier injectively, gives the stated whole-target bound, and continues to arbitrary old carriers by bounding only the complement; those are the features used in the remainder of the paper. The Lean corollary itself does not assume that the operation signature is finite.

Proposition 4.7 (Closure of the old image)**.** *Let* $\iota_D : A \hookrightarrow R(D)$ *be the canonical old embedding. Its image is closed under every total operation of* $R(D)$ *if and only if the old evaluator is total.*

Indeed, a defined old application reduces to an old value. If $\operatorname{eval}_D(f, a, b) = \operatorname{none}$, however, then

$$f^{R(D)}(\iota_D(a), \iota_D(b)) = \operatorname{susp}(f, \operatorname{old}(a), \operatorname{old}(b)),$$

which is not in the old image. Lean checks both the pointwise escape statement `ResolutionSemantics.ResidualComparison.undefinedApplicationEscapesOldImage` and the equivalence `ResolutionSemantics.ResidualComparison.generatedOldClosedIffEvaluatorTotal`.

The selected-subterm encoding together with an overflow state has familiar tree-automata antecedents [14]; it is not presented as a new automaton architecture. Its use here is constrained by compatibility with a pointwise-preserved partial base. The resulting targets are generally not finite: A may be infinite, while only the added tags and $\star$ are counted. A further computational analogy is the separating-words problem. Sublinear worst-case bounds were established by Goralčík and Koubek, Robson, and Chase [12, 19, 25]. Along a unary comb, a Resolution observer likewise iterates one transition map from a fixed starting state. This analogy confirms that our linear estimate is a baseline, but it does not transfer word bounds automatically: arbitrary Answer trees branch, and every observer must extend the pointwise-preserved partial algebra. The resulting relative separation complexity is open. Determining whether this exact observer class already has a standard name remains part of our specialist literature audit.

5 The Observational Filtration and Completion

Each $\equiv_n$ is an equivalence relation. The formal model family yields a descending filtration: agreement at a larger observational budget implies agreement at every smaller budget. By theorem 4.3,

$$\bigcap_{n \in \mathbb{N}} \equiv_n = \{(x, x) \mid x \in R(D)\}.$$

Thus the filtration is separated.

A symbolic non-Archimedean distance can be read from the least separating stage. Equal points have distance zero; a distinct pair receives the level of its least finite separator. The strong triangle law follows from transitivity of every stage relation. This is standard filtered-space machinery, analogous to established metric treatments of trees [1, 15].

A sequence (x_k) is Cauchy when, for every stage n, its tail is pairwise $\equiv_n$-equivalent. Two Cauchy sequences are asymptotically equivalent when they eventually agree at every fixed stage. Let

$$\widehat{R(D)} = \{\text{Cauchy sequences in } R(D)\}/\sim .$$

The constant-sequence map is injective by separatedness and stage-dense by the usual late-term argument. A diagonal construction proves completeness.

Theorem 5.1 (Finite-base properness criterion)**.** *Suppose the old carrier A admits an injection into* $\mathrm{Fin}(m)$ *for some m. Then the canonical embedding*

$$R(D) \longrightarrow \widehat{R(D)}$$

is surjective if and only if the old evaluator is total. Equivalently, this embedding is not surjective if and only if there exist $f \in \Sigma$ and $a, z \in A$ with $\mathrm{eval}_D(f, a, z) = \mathrm{none}$.

Proof. A stage-n observer has at most $m + n + 1$ states: the finitely coded old carrier, n fresh tags, and overflow. If $\mathrm{eval}_D(f, a, z)$ is undefined, let c_k be the left comb obtained by iterating this application. Along the comb family every observer acts by the orbit of one self-map. Hence $(c_{k!})_{k \in \mathbb{N}}$ is Cauchy: sufficiently late factorial indices have difference divisible by every possible orbit period.

For non-convergence, fix a proposed finite Answer x. Choose the preserving finite-tag observer constructed from the finite subterm closure of x together with a comb prefix longer than x. It encodes x and the prefix by distinct non-overflow states. The next comb transition matches no selected normalized node, so it enters overflow; no selected transition has overflow as an encoded child, making overflow absorbing along the rest of the comb. Thus this single finite stage separates x from every sufficiently late factorial comb. No generated Answer is a limit.

Conversely, if the old evaluator is total, structural induction shows that every generated Answer is old. Cauchy agreement at a fixed separating stage then makes every Cauchy sequence eventually constant, so the generated space is complete. Completeness is equivalent to surjectivity of the canonical completion embedding. The checked public statements are `ResolutionSemantics.FiniteBaseCompletion.completeIffTotal` and `ResolutionSemantics.FiniteBaseCompletion.embeddingNotSurjectiveIffExistsUndefined`. □

The one-point always-undefined algebra is the smallest corollary and still provides an explicit represented completion point through `ResolutionSemantics.OnePointCompletion.addsPoint`. The theorem does not by itself decide infinite-base examples. The following independent criterion does.

Theorem 5.2 (Old-fixing-context properness)**.** *Let D be any binary partial algebra; no finiteness assumption is made on its old carrier. Suppose*

$$\mathrm{eval}_D(g, a, b) = \mathrm{none} \quad \textit{and} \quad \mathrm{eval}_D(u, x, e) = \mathrm{some}(x) \qquad (x \in A)$$

for some $g, u \in \Sigma$ and $a, b, e \in A$. Define

$$c_0 = \mathrm{susp}(g, \mathrm{old}(a), \mathrm{old}(b)), \qquad c_{k+1} = \mathrm{susp}(u, c_k, \mathrm{old}(e)).$$

Then:

(a) *$c_1 \neq c_2$ but $c_1 \equiv_0 c_2$;*
(b) *for every m, the distinct pair $c_{(m+2)!}, c_{(m+3)!}$ agrees at stage m, and its least finite-tag separation rank is greater than m;*
(c) *$(c_{k!})_{k\in\mathbb{N}}$ is Cauchy and has no generated limit.*

Consequently the generated filtered space is not complete, the finite-tag separation ranks are unbounded, and the canonical completion embedding is not surjective.

Proof. Let T be a stage-m preserving observer and let $F(s) = u_T(s, \mathrm{old}(e))$. Preservation gives $F(\mathrm{old}(x)) = \mathrm{old}(x)$ for every $x \in A$. For this orbit, collapse the entire old carrier to one settled state while retaining the m fresh tags and overflow. The collapse commutes with F and has $m + 2$ states. Equality in the collapsed orbit lifts to the original orbit: complement states were retained injectively, while an orbit that reaches an old value $\mathrm{old}(x)$ stays at that exact value because $F(\mathrm{old}(x)) = \mathrm{old}(x)$. Finite-orbit periodicity therefore implies that sufficiently late indices whose difference is divisible by $(m + 2)!$ have equal T-values. Applying this simultaneously to factorial indices proves that $(c_{k!})$ is Cauchy and gives the stage-m agreement in clause (b). Constructor count proves the pair distinct. Since a distinct pair agrees at stage m, the exact rank characterization makes its least separating budget greater than m.

At stage zero the complement consists only of overflow. If F sends overflow to an old value, the next application fixes that value; if it sends overflow to overflow, the same table entry fixes it. Thus $F^2 = F$, proving $c_1 \equiv_0 c_2$, while constructor count again proves $c_1 \neq c_2$.

For non-convergence, fix a proposed finite Answer x. The standard candidate-tailored preserving observer encodes x together with a repeated $u(-, e)$ prefix longer than x. The next iterate is absent from the selected finite table and therefore enters overflow; overflow is absorbing along the remaining tail. The candidate remains encoded by a non-overflow state, so one finite stage separates x from every sufficiently late factorial term. Hence no generated Answer is a limit. The checked public statements are `ResolutionSemantics.OldFixingContextCompletion.factorialSequenceCauchy`, `ResolutionSemantics.OldFixingContextCompletion.factorialSequenceNoLimit`, `ResolutionSemantics.OldFixingContextCompletion.separationRankGreaterThanBudget`, and `ResolutionSemantics.OldFixingContextCompletion.embeddingNotSurjective`. □

Natural and integer arithmetic both satisfy the hypotheses with undefined seed $0/0$ and old-fixing context $x \mapsto x + 0$. Thus theorem 5.2 proves that both arithmetic completion embeddings are non-surjective and that neither filtration has a finite stage equal to equality. The direct public corollaries include `ResolutionSemantics.NatDivision.oldFixingCriterion`, `ResolutionSemantics.NatDivision.separationRanksUnbounded`, `ResolutionSemantics.IntDivision.completionEmbeddingNotSurjective`, and `ResolutionSemantics.IntDivision.separationRanksUnbounded`.

Proposition 5.3 (Incomparability of the properness hypotheses)**.** *Neither properness hypothesis subsumes the other.*

(i) *The one-point always-undefined algebra has a one-state code, an undefined old application, and a proper completion, but it admits no old-fixing-context witness.*
(ii) *Natural arithmetic has the old-fixing witness $x \mapsto x + 0$ and a proper completion, but its old carrier admits no finite code.*

Proof. For the one-point algebra, any old-fixing witness would require its sole operation to satisfy $\mathrm{eval}_D(u, \star, \star) = \mathrm{some}(\star)$, contradicting the definition $\mathrm{eval}_D \equiv \mathrm{none}$. Its one-state code, undefined operation, and proper completion are explicit.

For natural arithmetic, the witness is the undefined seed $0/0$ together with the context $x \mapsto x+0$. If a finite code of bound m existed, applying the bounded pigeonhole theorem to the $m+1$ inputs $0, \ldots, m$ would give distinct i, j with equal codes, contradicting injectivity. The checked statements are `ResolutionSemantics.PropernessCriteria.onePointProperWithoutOldFixing` and `ResolutionSemantics.PropernessCriteria.natProperWithoutFiniteCoding`. □

Proposition 5.4 (Completed operations)**.** *Every generated operation $f_{R(D)}$ is jointly stage-compatible and therefore extends pointwise to a unique jointly nonexpansive operation*

$$\widehat{f} : \widehat{R(D)} \times \widehat{R(D)} \longrightarrow \widehat{R(D)}.$$

For embedded points,

$$\widehat{f}(\eta x, \eta y) = \eta(f_{R(D)}(x, y)).$$

The completed carrier and operations form a compatible total extension of D with injective old embedding. Its completeness is exposed by `ResolutionSemantics.completionComplete`; the compatible algebra is `ResolutionSemantics.completedAlgebra`.

5.1 Relative universal property

The formalization defines a category-like class of targets consisting of:

- a compatible total extension of D with injective old embedding;
- a separated descending filtration;
- nonexpansive generated interpretation;

- jointly nonexpansive primitive operations;
- completeness of the filtered carrier.

For every such target T, generated interpretation extends uniquely from the stage-dense subalgebra, and binary-extension uniqueness shows that the extension preserves all primitive operations. Hence the completion is initial relative to this specified target class. The exact Lean package is `Resolution.External.FilteredTotalAlg.completedResolutionFilteredAlgebra_initial`.

The adjectives “initial,” “reflective,” and “idempotent” are mathematically correct only with these hypotheses. They are consequences of standard completion arguments and are not presented here as separate novelty claims.

6 Exact Equational Conservativity

Let algebraic terms contain variables, named old-carrier constants, and binary operation symbols. They can be evaluated in $R(D)$ and in $\widehat{R(D)}$. Embedding commutes with term evaluation by induction. Moreover, every term operation preserves each stage relation because the primitive operations do.

Fix a completed assignment ρ and a stage n. Stage-density allows us to choose, independently for every variable, a generated assignment ρ_n with

$$\rho(v) \equiv_n \eta(\rho_n(v)).$$

Stage compatibility then gives

$$[\![s]\!]_\rho \equiv_n \eta([\![s]\!]_{\rho_n})$$

and similarly for t.

Theorem 6.1 (Exact equation preservation). *For every constant-bearing equation $s = t$,*

$$\Big(\forall \rho : \mathrm{Var} \to R(D),\ [\![s]\!]_\rho = [\![t]\!]_\rho\Big)$$

if and only if

$$\Big(\forall \widehat{\rho} : \mathrm{Var} \to \widehat{R(D)},\ [\![s]\!]_{\widehat{\rho}} = [\![t]\!]_{\widehat{\rho}}\Big) .$$

Proof. For the forward implication, approximate an arbitrary completed assignment at every stage. The generated equation identifies the two embedded approximants, so the completed term values agree at every stage. Separatedness yields exact equality. The reverse implication follows by evaluating on embedded generated assignments and using injectivity of the embedding. □

The public theorem is `ResolutionSemantics.equationConservative`.

Remark 6.2 (Conditional equations). The theorem concerns universal equations. Arbitrary quasi-equations do not follow from density alone. The formal development transfers conditional laws only under explicit stage-robust or dense-premise hypotheses. No unrestricted quasi-equational conservativity claim is made.

7 Arithmetic Examples

The general construction does not depend on ring or field axioms. We include natural-number and integer instances because they make the behavior at a zero denominator concrete.

7.1 Natural numbers

Take operations $+$, $\cdot$, and $/$ on $\mathbb{N}$, with division undefined exactly when the denominator is zero. For $b \neq 0$,

$$\text{divide}(a, b) = \text{old}(a/b).$$

At zero,

$$\text{divide}(a, 0) = \text{susp}(\mathsf{div}, \text{old}(a), \text{old}(0)).$$

Therefore

$$a/0 = b/0 \Longrightarrow a = b,$$

and no $a/0$ is an old natural number. These properties remain true after the completion embedding. They are exposed by `ResolutionSemantics.NatDivision.singularFamilyInjective` and `ResolutionSemantics.NatDivision.singularFamilyDisjoint`.

The observational completion is nontrivial in a stronger sense. By theorem 5.2, repeatedly adding the old right identity to $0/0$ produces distinct finite Answers that no fixed observation stage can fully separate. The factorially sampled family is Cauchy without a generated limit, so the completion contains an explicit point outside the generated image, and the explicit pair at budget m has separation rank greater than m. This uses the old-fixing-context criterion rather than the finite-base theorem.

7.2 Integers

The integer instance additionally includes subtraction. Again, every nonzero division agrees with the old evaluator, while

$$a/0 = \text{susp}(\mathsf{div}, \text{old}(a), \text{old}(0)).$$

The completed singular family is injective, and completed $0/0$ is not any embedded integer. The paper-facing results are `ResolutionSemantics.IntDivision.singularFamilyInjective` and `ResolutionSemantics.IntDivision.zeroDivZeroNotOld`.

Integer addition also has old right identity zero. Hence the same old-fixing-context criterion proves that every finite integer-arithmetic stage is strictly coarser than equality, its separation ranks are unbounded, and its completion embedding is not surjective. These corollaries are `ResolutionSemantics.IntDivision.everyFiniteStageNotEquality`, `ResolutionSemantics.IntDivision.factorialOldFixingCauchy`, and `ResolutionSemantics.IntDivision.completionEmbeddingNotSurjective`.

> The construction does not assign an ordinary numerical value to $0/0$. It gives $0/0$ a structured Answer in a conservative extension.

This distinction matters when comparing with totalized arithmetic. In an involutive meadow one usually sets $0^{-1} = 0$; in a common meadow exceptional terms propagate one designated error; wheels adopt a different algebraic extension with infinity- and error-like behavior [3, 6, 11]. Resolution chooses provenance retention rather than a fixed old value or one undifferentiated error.

8 Relation to Prior Work

8.1 Free completion and partial algebras

The closest structural prior art is the free completion of partial algebras and the Schmidt construction [28]. Theorem 3.3 should be read as a Lean-checked concrete presentation of that general idea for the one-sorted binary setting. Other completion and encoding results show that partial-to-total extension is not in itself a novel contribution [7, 8, 18, 23]. In particular, section 2 follows Diaconescu's choices of Γ, α, β, and γ and checks them in the repository's concrete typed syntax. Its value here is the verified interface between that established encoding and Resolution normal forms, not a priority claim.

8.2 Ground terms, recognizability, and operation trees

Separating operation syntax from interpretation is standard in term algebras, free monads, and algebraic-effects semantics [24]. We do not claim the first operation-tree semantics. The specific normalization discipline is: reduce an old-defined node immediately and retain an old-undefined node with its already resolved children.

For finite bases and finite signatures this normalization is also a finite ground-equational presentation, as detailed in remark 4.6. Recognizable tree languages, syntactic congruences, and finite tree automata are therefore direct prior art for the ordinary residual-finiteness consequence [9, 14, 16, 20, 27]. The pair-dependent separator likewise uses the standard finite closure of selected subterms plus a sink-like state. The distinguishing constraint in this paper is not that combinatorial pattern but its compatibility with a possibly infinite base that is fixed pointwise while only the complement is budgeted.

8.3 Tree metrics and completion

Ultrametric and algebraic treatments of infinite trees long predate this work [1, 15]. Our completion is distinguished by its source of finite observations: agreement in all compatible finite-tag models, rather than prefix depth. The Cauchy-sequence quotient and continuous extension arguments are standard once the separated filtration is available.

The distinction is not a matter of presentation. Write $x \equiv_n^d y$ for agreement of two Answers on all levels strictly above depth n; this is the filtration whose completion is the classical infinite-tree space. The two uniformities differ, and a single family of witnesses shows it.

Proposition 8.1. *Let D be a binary partial algebra with* $\mathrm{eval}_D(f, a, z) = \mathrm{none}$ *for some f, a, z. Define the* combs *by $c_0 = a$ and $c_{k+1} = f(c_k, z)$. Then the sequence* $(\mathrm{Res}_D\, c_k)_{k\in\mathbb{N}}$ *is Cauchy for the filtration $\equiv_n^d$ but not Cauchy for the filtration $\equiv_n$.*

The formal statement is `Resolution.Probe.depthCauchy_not_observationalCauchy`. Each comb suspends at every level, so $\mathrm{Res}_D\, c_i$ and $\mathrm{Res}_D\, c_j$ agree above depth n whenever $i, j \geq n$: under the tree metric the sequence converges to the infinite comb. For the finite-tag filtration one preserving model with a *single* fresh tag suffices to break it. Send an undefined old pair to the tag, the tag to $\star$, and $\star$ back to the tag; the fold along the combs then alternates between the tag and $\star$, so adjacent combs are separated at stage 1 for every k, however deep the disagreement lies (`Resolution.Probe.comb_separatesAt_one`). A fixed observational budget therefore sees through unbounded depth, which prefix truncation cannot do. Consequently the identity on $R(D)$ does not carry $\equiv_n^d$-Cauchy sequences to $\equiv_n$-Cauchy ones, and no uniformly continuous isomorphism of the two completions commutes with the canonical embeddings. Only the statement about the two filtrations is formalized; the consequence for the completions is the standard argument on uniformly continuous maps and is not machine-checked.

In the arithmetic instance of section 7 the effect is sharper: the combs are the iterated singular divisions $(\cdots((a/0)/0)\cdots/0)$, and *no fresh tags at all* are needed. Sending each singular division to the successor of its numerator is a preserving model over $\mathbb{N} \sqcup \{\star\}$ that separates every pair of distinct combs at stage 0, letting the old carrier itself do the counting (`Resolution.Probe.NatProbe.natComb_pairwise_separated_at_zero`).

That stage-zero separator does not make the full natural-arithmetic filtration discrete. It exploits fresh old naturals generated by the iterated-division table. The different family $c_0 = 0/0$, $c_{k+1} = c_k + 0$ cannot use such tags: preservation forces $a + 0 = a$ on every old natural. In fact c_1 and c_2 agree in every stage-zero observer, and theorem 5.2 gives an indistinguishable distinct pair at every finite stage.

The iterated-division comb also bears on the sharpness of the bound in theorem 4.3. Since $|\mathrm{Res}_D\, c_k| = 2k + 1$, that bound permits $4k + 4$ tags to separate c_k from c_{k+1}, whereas one tag suffices in general and none in arithmetic. The linear budget is therefore far from tight on deep pairs; we do not know its exact order of growth.

8.4 Finite complements and profinite comparison

Finite-complement extensions are classical in semigroup theory under the name of finite Rees index. Cain and Maltcev's survey explicitly records that residual finiteness passes both to finite-Rees-index subsemigroups and to their finite-complement extensions, citing the theorem of Ruškuc and Thomas [10, 26]. Thus residual finiteness is not a gap in that literature. Miller, O'Reilly, Quick, and Ruškuc study residual, weak, strong, and complete separability conditions for semigroups in the finite-quotient tradition [21].

These results are important adjacent prior art, but neither citation by itself identifies the observer class used here. Our base is a pointwise-fixed partial subalgebra, while a separator is a compatible total extension whose complement over that base is finite; the separating codomain need not itself be finite when the base is infinite. Moreover, ordinary Rees index starts with a closed subsemigroup. By proposition 4.7, the old image in the generated Answer algebra is closed under the total operations exactly in the already total case. Hence finite Rees index is not literally the inclusion used by a genuinely partial Resolution algebra. The remaining specialist question is whether separation by finite-complement extensions of a pointwise-fixed *partial* base has already been studied under a generalized Rees-index, relative residual-finiteness, or separability formulation.

There is likewise a mature theory of profinite completions and natural extensions of residually finite algebras, as well as categorical formulations through profinite monads [13, 17]. Recent work on profinite trees gives a particularly close comparison on the tree side [22]. Those constructions are organized by finite quotients or finite recognizing algebras. Our stages instead allow a possibly infinite preserved base and bound only the genuinely new complement. The finite-base properness theorem above is therefore not presented as a theorem about ordinary profinite completion. The documented search has found these analogues, but no exact equivalence or prior statement of the criterion; specialist review is still required.

8.5 Totalized division

Meadows, common meadows, wheels, non-involutive meadows, partial meadows, and fracpair constructions provide mature alternatives with strong equational and initial-algebra results [2–6, 11]. Resolution is not proposed as uniformly superior. Its design goal is different: preserve old-defined calculations while retaining the provenance of old-undefined applications.

8.6 Evidence-qualified contribution

The established ingredients include free completion, operation-tree syntax, finite-signature ground-term recognizability, and standard Cauchy-completion machinery. Against that background, the package submitted for specialist assessment has three parts:

1. an intrinsic relative observer class that fixes an arbitrary, possibly infinite partial base pointwise and budgets only its complement, together with a canonical finite-tag presentation, pairwise separation, and an explicit base-faithful construction bound;
2. the induced observational completion, including an exact finite-base properness criterion, an independent arbitrary-base old-fixing-context criterion, their formal incomparability, the contrast with prefix-depth completion, and exact universal equational conservativity;
3. an audited Lean development comprising the many-sorted encoding bridge and the binary separation–completion results, with natural- and integer-arithmetic instances and explicit unbounded-rank witnesses.

Our search has not found this exact relative separation–completion package. That reports the present search result, not proof of historical priority or a claim that each constituent is new.

9 Formalization and Trust Boundary

The paper-only repository exposes a deliberately small public interface:

```
import ResolutionSemanticsCompletion
import ResolutionIntrinsicFiniteComplementAPI
import ResolutionFiniteBasePropernessPublic
import ResolutionOldFixingContextPropernessPublic
import ResolutionPropernessCriteriaComparison
import ResolutionArithmeticPropernessPublic
```

```
import ResolutionDiaconescuManySorted
import ResolutionDiaconescuContinuation

#check ResolutionSemantics.freeCompletionUniversal
#check ResolutionSemantics.OutsideOld
#check ResolutionSemantics.oldOrOutsideEquiv
#check ResolutionSemantics.outsideEquiv
#check ResolutionSemantics.qualitativeFiniteComplementSeparating_theorem
#check ResolutionSemantics.intrinsicExtensionHasCanonicalPresentation
#check ResolutionSemantics.finiteExternalSeparation
#check ResolutionSemantics.finiteSeparationBound
#check ResolutionSemantics.completionComplete
#check Resolution.External.FilteredTotalAlg.
    completedResolutionFilteredAlgebra_initial
#check ResolutionSemantics.equationConservative
#check ResolutionSemantics.FiniteBaseCompletion.completeIffTotal
#check ResolutionSemantics.FiniteBaseCompletion.
    embeddingNotSurjectiveIffExistsUndefined
#check ResolutionSemantics.OldFixingContextCompletion.embeddingNotSurjective
#check ResolutionSemantics.OldFixingContextCompletion.
    separationRankGreaterThanBudget
#check ResolutionSemantics.PropernessCriteria.onePointProperWithoutOldFixing
#check ResolutionSemantics.PropernessCriteria.natProperWithoutFiniteCoding
#check ResolutionSemantics.NatDivision.separationRanksUnbounded
#check ResolutionSemantics.IntDivision.completionEmbeddingNotSurjective
#check ResolutionSemantics.IntDivision.separationRanksUnbounded
#check Resolution.Diaconescu.ManySorted.term_evalPartial_sound
#check Resolution.Diaconescu.ManySorted.term_evalPartial_complete
#check Resolution.Diaconescu.ManySorted.alpha_satisfaction_condition
#check Resolution.Diaconescu.ManySorted.gamma_satisfiesGamma
#check Resolution.Diaconescu.ManySorted.betaGammaPartialAlgEquiv
#check Resolution.Diaconescu.ManySorted.gamma_persistent_liberality
#check Resolution.Diaconescu.ManySorted.betaHom_comp
#check Resolution.Diaconescu.ManySorted.gammaBetaHomEquiv
#check Resolution.Diaconescu.ManySorted.gammaHom_comp
#check Resolution.Diaconescu.ManySorted.Formula.sat_iff_of_equiv
#check Resolution.Diaconescu.ManySorted.semantic_consequence_equivalence
#check Resolution.Diaconescu.ManySorted.gammaHom_generatedOld
#check Resolution.Diaconescu.ManySorted.gammaHom_generatedTotal
#check Resolution.Diaconescu.ManySorted.gammaHom_generatedPartial
```

The repository pins Lean 4.33.0. The command

```
bash scripts/verify.sh
```

compiles the checked-in proof dependency chain, checks every manuscript-linked declaration, rejects direct proof-hole commands, and requires each audited public theorem to depend only on `propext`, `Classical.choice`, and `Quot.sound`. The separate command `bash scripts/build-paper.sh -check-committed` rebuilds this PDF deterministically and verifies that the checked-in copy matches. Finite indexing and witness selection use ordinary classical choice in documented places. The result is relative to Lean's underlying logic and implementation; no foundation-independent consistency claim is made.

The paper gives full mathematical arguments for the free-completion and finite separator constructions because those determine its positioning. Routine completion lemmas are summarized and linked to exact declarations in `THEOREM_MAP.md`.

10 Limitations and Open Questions

1. The Γ–α–β–γ bridge is many-sorted and supports arbitrary finite arities and separate TF/PF families. The finite-complement separation and observational-completion layer remains one-sorted and binary. Extending that second layer, and comparing it directly with the full Schmidt construction, remains open.
2. The bridge checks the concrete operations-only sentence and model constructions used here; it does not formalize institution morphisms, arbitrary signature morphisms, or the initial-specification corollaries of persistent liberality.
3. For finite bases and finite signatures, ordinary residual finiteness is already explained by classical ground-term recognizability. The precise relation of the arbitrary-base, pointwise-preserving finite-complement observer class to known relative residual-finiteness constructions remains under review.
4. The linear tag bound is a baseline upper bound, not an optimality theorem; its relative state complexity is open.
5. Properness is classified exactly for finitely coded old carriers. The old-fixing-context hypothesis is sufficient but not necessary, and neither of the two stated hypotheses subsumes the other; a full arbitrary-carrier classification remains open.
6. The construction is semantic, not an efficiency claim. Retaining full provenance can produce large finite trees.
7. No ordinary field law is asserted globally on singular Answers unless it is separately proved in the generated algebra.
8. Constant-bearing universal equations transfer exactly; conditional equations require additional hypotheses.
9. Historical priority for the combined package remains unresolved until specialist review is complete.

11 Conclusion

The many-sorted bridge places the project explicitly downstream of Diaconescu's total-algebra encoding: the concrete TF/PF Horn theory, the α–β satisfaction condition, and the canonical γ lift are all checked in Lean, while their mathematical priority remains with the established source. Resolution semantics is then best understood as a normalized structured presentation of a free compatible completion of a partial algebra. Its basic conservativity follows from the reduction rule. The substantive additional construction is intrinsic finite-complement separation over the pointwise-preserved old fragment. It separates every generated pair while adding only finitely many states beyond the base; a derived finite-tag normal form supplies the explicit witnesses and induces a separated observational filtration. For finitely coded bases this implies ordinary residual finiteness with an explicit whole-target bound; the qualitative finite-signature consequence also follows from classical ground-term recognizability. For genuinely partial algebras the old image is not a total subalgebra, so the finite-complement condition is not literally ordinary Rees index. The general linear estimate is a baseline; its sharper relative complexity is open. Over every finitely coded old carrier, the completion is proper exactly when an old application is undefined. This finite-base hypothesis and the old-fixing-context hypothesis are incomparable, as witnessed by the one-point algebra and natural arithmetic respectively. Standard completion methods then produce a complete total algebra with canonical operations and exactly the same universal equations in the language with named old-carrier constants.

The old-fixing-context theorem explains the arithmetic examples uniformly: for both natural and integer arithmetic every finite stage is strictly coarser than equality, the separation ranks are unbounded, and the factorial repeated-$+0$ family supplies an explicit completion point outside the generated image. At the same time, $0/0$ remains a structured element outside the old numerical image; the construction does not replace ordinary arithmetic with a new numerical convention. The formal results are reproducible in Lean, while the novelty claim is intentionally limited to the specific verified combination described above.

References

[1] A. Arnold and M. Nivat. The metric space of infinite trees: Algebraic and topological properties. *Fundamenta Informaticae*, 3(4):445–475, 1980. doi: 10.3233/FI-1980-3405.

[2] J. A. Bergstra and C. A. Middelburg. Division by zero in non-involutive meadows. *Journal of Applied Logic*, 13(1):1–12, 2015. doi: 10.1016/j.jal.2014.10.001.

[3] J. A. Bergstra and A. Ponse. Division by zero in common meadows, 2014. URL https://arxiv.org/abs/1406.6878. Version 4, revised 2021.

[4] J. A. Bergstra and A. Ponse. Fracpairs and fractions over a reduced commutative ring. *Indagationes Mathematicae*, 27(3):727–748, 2016. doi: 10.1016/j.indag.2016.01.007.

[5] J. A. Bergstra and J. V. Tucker. The rational numbers as an abstract data type. *Journal of the ACM*, 54(2):7:1–7:25, 2007. doi: 10.1145/1219092.1219095.

[6] J. A. Bergstra, Y. Hirshfeld, and J. V. Tucker. Meadows and the equational specification of division. *Theoretical Computer Science*, 410(12–13):1261–1271, 2009. doi: 10.1016/j.tcs.2008.12.015.

[7] I. Bethke, J. W. Klop, and R. C. de Vrijer. Completing partial combinatory algebras with unique head-normal forms. In *Proceedings of the Eleventh Annual IEEE Symposium on Logic in Computer Science*, pages 448–454. IEEE Computer Society Press, 1996. doi: 10.1109/LICS.1996.561461.

[8] I. Bethke, J. W. Klop, and R. C. de Vrijer. Extending partial combinatory algebras. *Mathematical Structures in Computer Science*, 9(4):483–505, 1999. doi: 10.1017/S0960129599002832.

[9] W. S. Brainerd. Tree generating regular systems. *Information and Control*, 14(2):217–231, 1969. doi: 10.1016/S0019-9958(69)90065-5.

[10] A. J. Cain and V. Maltcev. For a few elements more: A survey of finite Rees index, 2013. URL https://arxiv.org/abs/1307.8259.

[11] J. Carlström. Wheels—on division by zero. *Mathematical Structures in Computer Science*, 14(1):143–184, 2004. doi: 10.1017/S0960129503004110.

[12] Z. Chase. A new upper bound for separating words, 2020. URL https://arxiv.org/abs/2007.12097.

[13] L.-T. Chen, J. Adámek, S. Milius, and H. Urbat. Profinite monads, profinite equations, and Reiterman's theorem. In *Foundations of Software Science and Computation Structures*, volume 9634 of *Lecture Notes in Computer Science*, pages 531–547. Springer, 2016. doi: 10.1007/978-3-662-49630-5_31.

[14] H. Comon, M. Dauchet, R. Gilleron, F. Jacquemard, D. Lugiez, S. Tison, and M. Tommasi. Tree automata techniques and applications, 2008. URL https://www.irif.fr/users/tata/. Electronic monograph.

[15] B. Courcelle. Fundamental properties of infinite trees. *Theoretical Computer Science*, 25(2):95–169, 1983. doi: 10.1016/0304-3975(83)90059-2.

[16] M. Dauchet and S. Tison. The theory of ground rewrite systems is decidable. In *Proceedings of the Fifth Annual IEEE Symposium on Logic in Computer Science*, pages 242–248. IEEE Computer Society Press, 1990. doi: 10.1109/LICS.1990.113750.

[17] B. A. Davey, M. J. Gouveia, M. Haviar, and H. A. Priestley. Natural extensions and profinite completions of algebras. *Algebra Universalis*, 66:205–241, 2011. doi: 10.1007/s00012-011-0155-y.

[18] R. Diaconescu. An encoding of partial algebras as total algebras. *Information Processing Letters*, 109(23–24):1245–1251, 2009. doi: 10.1016/j.ipl.2009.09.008.

[19] P. Goralčík and V. Koubek. On discerning words by automata. In *Automata, Languages and Programming: 13th International Colloquium, ICALP 1986*, volume 226 of *Lecture Notes in Computer Science*, pages 116–122. Springer, 1986. doi: 10.1007/3-540-16761-7_61.

[20] D. Kozen. Complexity of finitely presented algebras. In *Proceedings of the Ninth Annual ACM Symposium on Theory of Computing*, pages 164–177. ACM, 1977. doi: 10.1145/800105.803406.

[21] C. Miller, G. O'Reilly, M. Quick, and N. Ruškuc. On separability finiteness conditions in semigroups. *Journal of the Australian Mathematical Society*, 113(3):402–430, 2022. doi: 10.1017/S1446788721000124.

[22] V. Moreau. Profinite trees, through Lawvere theories and the lambda-calculus, 2024. URL https://arxiv.org/abs/2402.13086. Version 2, revised 2026.

[23] G. A. R. Nesin. Completing partial algebra models of term rewriting systems. Master of logic thesis, Institute for Logic, Language and Computation, University of Amsterdam, 2011.

[24] G. D. Plotkin and J. Power. Algebraic operations and generic effects. *Applied Categorical Structures*, 11(1):69–94, 2003. doi: 10.1023/A:1023064908962.

[25] J. M. Robson. Separating strings with small automata. *Information Processing Letters*, 30(4):209–214, 1989. doi: 10.1016/0020-0190(89)90215-9.

[26] N. Ruškuc and R. M. Thomas. Syntactic and Rees indices of subsemigroups. *Journal of Algebra*, 205(2):435–450, 1998. doi: 10.1006/jabr.1997.7392.

[27] J. C. Vidal and E. C. Llópez. Congruence based proofs of the recognizability theorems for free many-sorted algebras. *Journal of Logic and Computation*, 30(2):561–633, 2020. doi: 10.1093/logcom/exz032.

[28] J. C. Vidal and E. C. Llópez. Functoriality of the Schmidt construction. *Logic Journal of the IGPL*, 31(5):822–893, 2023. doi: 10.1093/jigpal/jzac048.